

Harmanan Gurvinder Kohli

AI Systems Engineer | Machine Learning Engineer

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Research Interests

Large Language Model Systems • Agentic Workflows • Retrieval-Augmented Generation • Multimodal AI • AI Systems • Explainable AI • Semantic Retrieval • Memory-Augmented Agents • Tool-Using AI Systems • AI Systems Evaluation

Education

Birla Institute of Technology and Science (BITS), Pilani

M.Tech. Data Science and Engineering

Apr 2022 – May 2024

CGPA: 8.14/10

Relevant Coursework: Machine Learning, Deep Learning, Natural Language Processing, Data Mining, Data Structures and Algorithms

SRM Institute of Science and Technology

B.Tech. Computer Science and Engineering

Jul 2014 – May 2018

CGPA: 8.14/10

Relevant Coursework: AI & Expert Systems, Database Management Systems, Software Engineering, Data Mining, DSA

Research Experience

Master's Dissertation: Explainable AI for Predicting Malware Infection

Graduate Researcher

Nov 2023 – Apr 2024

BITS Pilani

- Investigated interpretable machine learning techniques for large-scale malware prediction using Microsoft's Malware dataset containing 8.9 million devices.
- Developed predictive models achieving 66.7% accuracy and 66.4% recall while comparing Explainable Boosting Machines (EBM), SHAP and LIME for local and global model explanations.
- Studied explainability methods to improve stakeholder trust, model transparency and GDPR-aligned decision support for cybersecurity applications.
- Engineered memory-efficient preprocessing pipelines for a 4.1 GB dataset using Polars, reducing preprocessing memory consumption by nearly 50%.

Launchpad.ai (via Fellowship.ai)

Data Science Fellow

Jan 2024 – Mar 2024

- Developed and evaluated a multimodal product pricing system integrating Gemini Pro Vision, Retrieval-Augmented Generation (RAG), VectorDB retrieval, and web search for product valuation.
- Designed hybrid retrieval workflows combining vector similarity search with external web intelligence to improve robustness for previously unseen products.
- Implemented scalable retrieval pipelines supporting more than 100K product records while reducing manual product lookup effort.
- Built automated Python and Selenium data collection pipelines for improving retrieval results.

Personal Projects

FinSight: Enterprise Multi-Agent Investment Intelligence Platform

Independent Personal Project

May 2026 – Present

- Architected a multi-agent AI platform for investment research, orchestrating specialized agents for retrieval, quantitative analysis, market intelligence, analytics and response validation.

- Designed a framework-agnostic agent architecture using Google ADK, LangGraph, LangChain, LlamaIndex, CrewAI, PydanticAI, OpenAI Agents SDK and the Agent-to-Agent (A2A) Protocol to study interoperable agent collaboration.
- Implemented a unified Model Context Protocol (MCP) server exposing structured financial data, SEC filings, and external knowledge sources.
- Implemented hybrid Retrieval-Augmented Generation architecture combining dense retrieval, BM25 search, reranking, metadata filtering, and persistent conversational memory to improve factual grounding.
- Designed evaluation and observability pipelines using RAGAS and Langfuse to analyze retrieval quality, reasoning performance, hallucinations, and end-to-end system behaviour.

Industry Experience

Capgemini Invent

GenAI Consultant / AI Systems Engineer

Aug 2025 – Present

Mumbai, India

- Designed an agentic document intelligence system for long-context financial reasoning using semantic chunking, Retrieval-Augmented Generation and AWS Bedrock.
- Designed a domain-aware retrieval architecture with dedicated vector indexes for multiple financial document types, improving retrieval precision and response quality.
- Implemented deterministic multi-agent orchestration using LangGraph for document routing, retrieval, reasoning, and response generation with production-grade governance.
- Developed hybrid AI pipelines integrating Text-to-SQL with semantic retrieval on unstructured documents.
- Built an internal multimodal product intelligence accelerator comprising five specialized AI agents for pricing estimation, GTM strategy generation, IP/Copyright risk assessment, creative advertisement generation and conversational reasoning using vector retrieval and live web search.

Sterlite Technologies Ltd.

Data Scientist

Oct 2024 – Apr 2025

Mumbai, India

- Developed statistical safety-stock optimization models using demand forecasting and service-level optimization to improve inventory planning.
- Applied Explainable AI techniques including Counterfactual Explanations to improve transparency of RFQ classification models.
- Built multimodal document understanding solutions using Gemini Pro Vision for automated invoice extraction and structured financial data generation.

Tata Consultancy Services

Systems Engineer

Jan 2020 – Jan 2024

Mumbai, India

- Delivered enterprise analytics and planning solutions across SAP ecosystems supporting reporting, forecasting, and operational decision-making.
- Optimized enterprise data-processing workflows, reducing execution time from days to minutes through pipeline redesign and automation.
- Integrated business, financial, and geospatial datasets across SAP S/4HANA and GIS to enable large-scale analytics and reporting.

Open Source Contributions

Scikit-Learn

Open Source Contributor

- Contributed official documentation examples for `f_regression()` and `silhouette_score()` (PR #28104, PR #28125).

The Hundred-Page Language Models Book

Acknowledged Contributor

- Recognized as a contributor for improving open educational resources on Large Language Models.

Technical Skills

Programming Languages: Python, SQL

Machine Learning: PyTorch, TensorFlow, Scikit-Learn, XGBoost

AI Systems: Large Language Models, Retrieval-Augmented Generation (RAG), Agentic AI, AI Agents, LangGraph, LangChain, LlamaIndex, CrewAI, Prompt Engineering

Multimodal AI: Vision-Language Models, Multimodal Retrieval

Retrieval & Search: Vector Databases, BM25, Hybrid Retrieval, Reranking

Evaluation: RAGAS, Langfuse

Cloud & Infrastructure: AWS Bedrock, Docker, Git, REST APIs

Awards

- Best Team Award (2x), Tata Consultancy Services
- On-the-Spot Award (5x), Tata Consultancy Services
- Special Initiative Award, Tata Consultancy Services
- TechGig Code Gladiators 2018 – Semi-finalist